

Name of project :

YAFURAH II – Control System - SW

Factory Acceptance Test (FAT)

The FAT shall be performed by the vendor and will include the following activities (Test schedule):

Item	Description	Remark
1	Initial meeting (FAT document and plan review, scope of tests, schedule, etc.)	
2	Vendor documentation (including in-house test reports) check	
3	Hardware and software inventory check	not applicable in this part
4	Mechanical inspection	not applicable in this part
5	Wiring and termination inspection	not applicable in this part
6	Start-up test and general system functions	
7	System alarm test	not applicable in this part
8	Hardware redundancy and diagnostic	not applicable in this part
9	Operator interface	
10	IO tests from field wiring terminals to I/O terminals, to controller and including HMI.	not applicable in this part
11	Complex functionality and operation modes (for example, batch, sequence control)	
12	Test of communication links to subsystems and third-party interfaces	not applicable in this part
13	FAT rework, punch list development	
14	FAT close-out meeting	

NA – not applicable in this part

DSPK : Roman [Signature]
AUTEL : PAVEL FUKALA [Signature]

OBO Dooran
n 3 -07- 2026
OK [Signature]

FAT test report

Item 2 - Vendor documentation check

Purpose

To review all FAT relevant documents.

Item	Verify document	Test result	Remarks
1		☐ P ☐ F ☐ NA	
2	Manuals	☐ P ☐ F ☒ NA	Part of HW FAT
3	Quality documentation	☐ P ☐ F ☒ NA	Part of HW FAT

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

Signature:

OSPW
AUTEL

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03-07-2026
OK

Item 6 - Start-up test and general system functions

Purpose

To verify that the system has no problems to start up, recover from a power failure and perform on-line loading. Furthermore, it shall be verified that the system is operating within the given limits.

Reference documents

- Relevant vendor documentation
- Requirement specification (specified system limits)

Item	Description	Test result
1	New start (start from zero point ^a , stop/start)	☒ P ☐ F ☐ NA
2	On-line changes	☒ P ☐ F ☐ NA
3	Controller cycle time	☒ P ☐ F ☐ NA
4	Display call-up time	☒ P ☐ F ☐ NA
5	Value update time	☒ P ☐ F ☐ NA
6	System load (memory capacity, storage capacity, etc.)	☒ P ☐ F ☐ NA
7	Log-on strategy and level	☒ P ☐ F ☐ NA
8	Alarm processing strategy and acknowledgement	☒ P ☐ F ☐ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

Signature:

DSPK: *[Signature]*
AUTEL *[Signature]*

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(W) OK
03-07-2026
[Signature]

Item 9 - Operator interface

Purpose

To verify the functionality of standard and graphic display arrangements in accordance with the specification.

Reference documents

- Relevant vendor documentation
- HMI specification

Note

This item covers the static part of the display only. The dynamic part will be tested later, together with the configuration functions.

Recommendation: The graphic design shall be agreed upon with the end-user well in advance

Item	Description	Test result
1	Colors of background and color changes	☒ P ☐ F ☐ NA
2	Symbols	☒ P ☐ F ☐ NA
3	Static text and dynamic changes	☒ P ☐ F ☐ NA
4	Organization (jumps, transitions, sub pictures)	☒ P ☐ F ☐ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

Signature:

DEPW: *[Signature]*
AUTEL: *[Signature]*

OK
03-07-2026
[Signature]

Item 11 - Complex functionality and operation modes

Purpose

To verify the functionality of the system according to the given documents.

Reference documents

Examples of project relevant documents are:

- Functional description
- Function or logic diagram
- Cause & effect matrix (C&E)
- Pre-defined typical

Note

Complex in this context means large sequence controls, recipes, advanced controls, meshed loops, etc.

Check of individual function groups functionality eg. Autostart, autostop, release conditions, backup etc. Including HMI design.

Complete list of Function Groups on next page.

Item	Description	Test result
1	Check of functionality in detail with the related alarms, messages, display, trends, signal updating on graphics and face plates	☒ P ☐ F ☐ NA

Comments

Nonconformity to be recorded in the punch list, categorized and treated as agreed in the start-up meeting.

P = Pass

F = Fail

NA = Not applicable

Signature:

DSPK: *[Signature]*
AUTEL: *[Signature]*

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[Signature] OK

List of Function Groups and HMI for Checking:

Item	Function Group	Test result	Remarks
1	Protections	OK	1.4. 26
2	Turbine Control	OK	1.4. - 2.4. 26
3	Lubrication Oil System	OK	30.6. 26
4	Turning Gear System & Jacking Oil System	OK	30.6. 26
5	Control Oil System (High pressure hydraulic system)	OK	30.6. 26
6	Gland Steam System	OK	30.6. 26
7	Draining System including warming	OK	1.4. 26
8	AVR, Generator Synchronization and Protection System	OK	2.4. 26
9	Flash Tank	OK	1.4. 26
10	Counters	OK	2.4. 26
11	Turbostep (ATS)	OK.	2.4. 26

DSPW: Kung
AUTEL: Tan

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(W) OK
03-07-2026

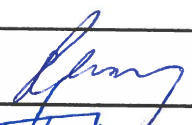
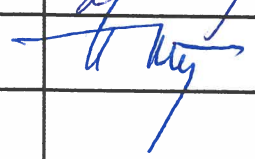
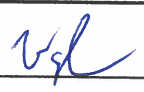
FAT certificate

☒ ACCEPTED

☐ NOT ACCEPTED

CUSTOMER	DOOSAN ŠKODA POWER s.r.o. Tylova 1/57, 301 28 Plzeň, Česká republika		
PROJECT	YAFURAH II	PROJECT NO	P100325
PLANT/UNIT	Harmaliyah GOSP 1, Kingdom of Saudi Arabia		
VENUE OF FAT	Autel, a.s. Oldřichovice 790, 739 01, Třinec, Česká republika	FAT FINISHED ON	

PERSON IN CHARGE/ Signatures:

CUSTOMER	Roman Filla DSPW	DEP.	
CUSTOMER	Miroslav Ardlidka OBO Doosan	DEP.	
CUSTOMER		DEP.	
VENDOR	PAVEL FUKALA AUTEL	DEP.	
VENDOR	PETR FUKALA AUTEL	DEP.	
VENDOR	JOSEF WAWRZYK AUTEL	DEP.	

ROMAN FILLA AUTEL
~~MIROSLAV ARDLIDKA~~
Filla DSPW




03-07-2026
OK
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FAT PUNCH LIST

Any incomplete work or nonconformities shall be recorded on the FAT punch list and categorized as follows:

- **Immediate** – FAT to be paused for rectification, FAT to continue after rectification;
- **On-going** – Rectification during FAT, will be completely rectified by end of FAT (typically for multiple instances of the same issue, each of which will be rectified as encountered and by the end of the FAT, all will be rectified. Rectification is performed before test. If later re-test is required, the item is classified as "Repeat");
- **Repeat** – FAT to be repeated for this item after rectification and before completion of FAT;
- **Before shipment** – To be rectified after FAT but before the system is shipped to the site, re-test, if any, to be documented and conducted during SAT;
- **At site** – Remaining work to be rectified after shipment, i.e., at site, and re-test, if any, to be documented and conducted during SAT or SIT; or
- **Future** – To capture issues not in the project or FAT scope but something to be addressed in the future.

Note

NO.	Reference to the applicable specification	Description of the issue	Description of the resolution and date completed	Acceptance (signature of buyer & vendor)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Kenny *John*

OK

03-07-2028
[Signature] OBO DOUVA

Document Type	Documentation Name	Documentation description	Language of Final documentation	Name of the electronic file and format	Document for final client
Contract	Contract/PO_Annexes	Contractual annexes with Technical specification and other requirements.	EN		
Counter obligation - documentation submitted by the Purchaser					
CP05	Basic for application software (logic diagrams with database of SW signals; referring to I11, I13)	Technical data according to the Technical Specification in annex 1.1 of the contract.	EN		
CP07	HMI design (referring to I16)	Technical data according to the Technical Specification in annex 1.1 of the contract.	EN		
Documentation for Design					
FP	CS_FAT_program	Detailed Function Approval Test program, all tests must be listed out.	EN ..	KKS of the main Equipment from 'Ts on the beginning of the file name; PDF	
I11	CS_Signal_transfer_list_TCS_DCS	Technical data according to the Technical Specification in annex 1.1 of the contract.	EN ..	I11_CS_Signal_transfer_list_TCS_DCS; PDF, EXCEL	JCP2-YR00-20MAA-240253
I13	CS_Logic_diagrams_for_control_function_groups	Technical data according to the Technical Specification in annex 1.1 of the contract. Please indicate on the cover sheet the name of the equipment (HMI design of Turbine Control System).	EN ..	I13_CS_Logic_diagrams_for_control_function_groups; PDF	JCP2-YF00-20MAA-240255
I16	CS_HMI_design	Technical data according to the Technical Specification in annex 1.1 of the contract. Please indicate on the cover sheet the name of the equipment (HMI design of Turbine Control System).	EN ..	I16_CS_HMI_design; PDF	JCP2-YL01-20MAA-240256
AM	AT_Assembly_manual	Assembly manual of the device according to Assembly Manual Dossier in annex 1.7.2 or annex 1.7.3.	EN ..	Documentation project number assigned by DSPW; PDF	JCP2-DM02-20MAA-270262
OM	CS_Operating_Maintenance_Manual	Operating and Maintenance Manual of the device according to Documentation Requirements for Operating and Maintenance instruction in annex 1.7.2 or 1.7.3.	EN ..	Documentation project number assigned by DSPW; PDF	JCP2-CQ02-20MAA-270263
CM	CS_Commissioning_manual	Commissioning manual of the device. Detailed requirements are described in annex 1.7.2 Documentation of commissioning tests. As the ITP is handed over separately in NOC3_OC_SAT for Assembly commissioning.	EN ..	Documentation project number assigned by DSPW; PDF	JCP2-CN00-20MAA-270264
TP	CS_Training_program	Will be filled in the template sheet Template - Training program in annex 1.7.2 or according to specification in annex 1.7.3.	EN ..	"Equipment name"; Training_program; Excel	JCP2-CQ02-20MAA-270265
TM	CS_Training_manual	Training manual must be prepared to such extent and quality that it would be possible to train the operating personnel according to it and Training program.	EN ..	"Equipment name"; Training_manual; Preferred Powerpoint presentation	JCP2-CQ02-20MAA-270266
Documentation for transport					
Progress report					

DSPW: Jimmy
AUTEL: Zia

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OBO DOORAN
03-07-2026

FOR INFORMATION

0	23.04.2026	Šíma	Havránek	Klečatský	First issue
Rev	Date	Prepared	Checked	Approved	Details of Revision

Owner



Owner's Engineer



Contractor



Sub-Supplier

Project

JAFURAH COGENERATION PROJECT – PHASE II EXPANSION

Title

Logic diagrams for control of functional groups

AUTEL : *filw*
DSPK : *fumy*

OK

03-07-2026

Document No.

JCP2-YF00-20MAA-240087

Rev.

0

Page-No.

405

List of logic diagrams:

03-07-2026
W

1. STG CONTROLLER
2. STG PROTECTIONS
3. LUBRICATION OIL
4. HPU (HIGH PRESSURE HYDRAULIC)
5. DRAINING SYSTEM
6. FLASH TANK
7. TURNING GEAR
8. GLAND STEAM
9. ATS (AUTOMATIC TURBINE SEQUENCE)
10. COUNTERS

Tested 30.6 – 3.7. 2026

DSPK : Roman Šima

AUTEL : PAVEL FUKALA

OBO Doosan : Miroslav Hrdlička

OK

Long

Fuk

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